



Fluid inertia controls mineral precipitation and clogging in pore to network-scale flows

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Mineral precipitation caused by fluid mixing presents complex control and predictability challenges in a variety of natural and engineering processes, including carbon mineralization, geothermal energy, and microfluidics. Precipitation dynamics, particularly under the influence of fluid flow, remain poorly understood. Combining microfluidic experiments and three-dimensional reactive transport simulations, we demonstrate that fluid inertia controls mineral precipitation and clogging at flow intersections, even in laminar flows. We observe distinct precipitation regimes as a function of Reynolds number (Re). At low Reynolds numbers ($Re < 10$), precipitates form a thin, dense layer along the mixing interface, which shuts precipitation off, while at high Reynolds numbers ($Re > 50$), strong three-dimensional flows significantly enhance precipitation over the entire intersection, resulting in rapid clogging. When injection rates from two inlets are uneven, flow symmetry-breaking leads to unexpected flow bifurcation phenomena, which result in enhanced concurrent precipitation in both downstream channels. Finally, we extend our findings to rough channel networks and demonstrate that the identified inertial effects on precipitation at the intersection scale are also present and even more dramatic at the network scale. This study sheds light on the fundamental mechanisms underlying mixing-induced mineral precipitation and provides a framework for designing and optimizing processes involving mineral precipitation.

fluid inertia | mixing-induced precipitation | clogging | flow intersection | microfluidics

Mineral precipitation is a phenomenon where solid phase minerals form from supersaturated solution, and it is critical to various natural and engineering processes, such as carbon mineralization, geothermal energy, critical mineral extraction, shale oil and gas recovery, membrane filtration, microfluidics, riverbank filtration, and more (1–17). Mineral precipitation has dichotomous outcomes, providing benefits in scenarios like carbon mineralization by stabilizing carbon in solid form (8, 9, 18), while posing challenges in scenes such as geothermal energy extraction and shale oil and gas recovery by obstructing flow pathways and decreasing system efficiency through clogging (7, 14, 17, 19). The initiation of precipitation hinges on the formation of supersaturated solutions, typically achieved when fluids with differing compositions are mixed. This fluid mixing is a pore-scale process that is known to be strongly affected by hydrodynamic conditions (20–24). However, most previous studies on mixing-induced precipitation have been performed with well-mixed batch reactors that neglect fluid flow effects (25, 26).

Subsurface systems commonly contain fractures and conduits that are highways for fluid flow (27, 28). Typical flow velocities in these systems are sufficiently fast to create inertial flows which manifest complex flow topologies (29–33). However, due to the common perception that porous media and microchannel flows are slow, the effects of fluid inertia are often overlooked in studies of mixing and reaction. Even in laminar flow regimes, it is known that fluid inertia can dramatically alter the flow topology and thus significantly affect flow and transport dynamics in channels and porous media (12, 34–41). In particular, three-dimensional (3D) flow structures are increasingly known to control mixing and reaction in flow channels (31), but previous studies of pore-scale mineral precipitation have focused on the Stokes flow regime and ignored inertia or 3D flow effects (42–46).

In natural fractured systems, flow intersections often serve as mixing and reaction hot spots, as they are sites where fluids with different properties can vigorously mix and react. Indeed, a recent field study demonstrated that fracture intersections can be hot spots for biogeochemical reactions because oxic and anoxic waters can interact at these locations (47). Further, a recent pore-scale study showed that 3D vortices formed at flow intersections can create hot spots for mixing-induced reactions within the fluid phase (31). In contrast to reactions within solutions, mineral precipitation is typically a heterogeneous process that relies on the availability of surface areas that serve as nucleation sites. Thus, the rate of

Significance

Mixing-induced mineral precipitation is a geological and chemical process in which dissolved mineral ions in liquid solutions are brought together by fluid flow to form solid mineral crystals. Mineral precipitation occurs under flow conditions in a variety of natural and engineered settings, including caves, aquifers, oceans, mining, and wastewater treatment. This study focuses on unraveling the intricate and poorly understood influence of hydrodynamic conditions on mineral precipitation. Through laboratory experiments and numerical simulations, we elucidate how fluid flow governs the spatiotemporal dynamics of precipitation in flow networks. Channels rapidly clog when subjected to fast flows, in stark contrast to their unclogged state under slower flows. This research bridges a critical knowledge gap and provides a framework for controlling mineral precipitation.

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precipitation is significantly influenced by how readily the reactants can access these sites (48). However, the impact of fluid inertia on mixing-induced precipitation at flow channel intersections remains unknown. Further, the role of pore-scale inertial effects on network-scale mineral precipitation is a larger open question.

In this study, we combined microfluidic experiments and 3D reactive transport simulations to uncover how fluid inertia controls mixing-induced precipitation and clogging at flow intersections and networks. Our experimental results show that fluid inertia dramatically alters spatiotemporal precipitation patterns. When inertial effects are negligible, we observed the formation of thin, dense mixing barriers that shut off precipitation, while when fluid inertia is significant, rapid clogging of intersections can occur. Our simulation results demonstrate fluid inertia results in the emergence of 3D flow topologies, which explains the dramatic change in the precipitation pattern. When the injection rates from the two inlets differ, flow symmetry-breaking results in unexpected 3D flow bifurcations that significantly enhance precipitation in both downstream channels. Finally, we find that our observations hold true and are even amplified in both rough channel intersections and channel networks. These results underscore the universality of our findings across scales and geometries.

Results

Inertia Effects Fundamentally Change Precipitation Patterns.

We investigated mixing-induced precipitation using a simple but relevant precipitation reaction: barite formation from barium chloride and sodium sulfate. This particular precipitation reaction, symbolized by the chemical equation $\text{BaCl}_2(aq) + \text{Na}_2\text{SO}_4(aq) \rightarrow \text{BaSO}_4(s) + 2\text{NaCl}(aq)$, is commonly encountered in various processes, including hydraulic fracturing, shale oil and gas recovery, and geothermal energy extraction (49–54). To better understand the fundamental physics that governs mineral precipitation, we first adopt a symmetric sinusoidal channel geometry, which captures key features of common flow channels, such as the variability in flow cross-sectional area and waviness. Barium chloride and sodium sulfate solutions, both with a concentration of 1.5 mM, are separately injected from two inlets, meet at an orthogonal intersection, and exit through two outlets (see *Materials and Methods* for details). We first investigate equal injection rate conditions and then uneven injection rate conditions. This is because uneven flow rates are often encountered in natural and engineering flow intersections. At the barium and sulfate concentrations used, we did not observe any indication of homogeneous nucleation in bulk solutions, and precipitation occurred primarily by heterogeneous precipitation on channel surfaces. The homogeneous and heterogeneous nucleation rates calculated with classical nucleation theory (CNT) also indicate the dominance of heterogeneous nucleation in our system (*SI Appendix*, Fig. S2). The fluid inertia is quantified by using Reynolds number, $\text{Re} = \rho u D_H / \mu$, where ρ is the density of fluids, u is the average velocity, D_H is the average hydraulic diameter of the channel, and μ is the viscosity of fluids. We vary Re from the O(1) to O(100) to cover creeping to inertial laminar flow regime. In fractured media, the flow generally falls within the advection-dominated regime, where flow velocities can reach as high as 1 m/s (32, 55, 56). Additionally, fracture apertures in these media typically vary between 0.005 and 0.1 cm (57, 58). Given these parameters, the Re in such settings can readily reach the order of 100. Additionally, in microfluidic applications involving mixing, particle separation, and focusing, similar Reynolds number ranges from the order of 1 to 100 are common (11–13). Thus, the range

of Re investigated in this study is relevant to flow in fractured media and microfluidics.

Fig. 1 shows grayscale images (main figures) of barite precipitation patterns under equal injection rates after 20 mL injection volume, alongside *Insets* depicting results from numerical simulations. The experimental results show that barite precipitation patterns change dramatically with increasing Re. When Re is 10, a dense, thin layer of barite precipitates gradually formed along the mixing interface at the intersection and extended from the surface into the channel interior (Fig. 1A). Increasing to Re = 30, the extent of the precipitates along the mixing interface and downstream channels become much wider (Fig. 1B). As the Re further increases (Re > 50), the precipitation becomes localized at the intersection and covers the entire intersection area (Fig. 1C–F). *Movies S1* and *S2* shows full precipitation dynamics at different Re values in time. We confirmed that the observed precipitation patterns at intersections are not limited to the sinusoidal geometry and are general. Similar patterns are observed in both straight and rough channel intersections (refer to *SI Appendix*, Figs. S3 and S14).

Hydrodynamic conditions in themselves have no effect on the fundamental thermodynamics and kinetics of the nucleation process, according to CNT (59). However, it can affect mineral precipitation by influencing the concentrations of reactants at the surfaces. Péclet number ($Pe = u D_H / D$, where D is the diffusion coefficient) is much larger than one at Re = 10 and further increases as Re increases. As solute transport is dominated by advection, the concentrations of reactants at the surfaces should be sensitive to the changes in flow topology. Thus, changes in the flow topology from increases in Re are likely to explain the changing patterns of precipitation. Since mixing and surface accessibility are the main factors constraining precipitation in this system, the changing precipitation dynamics should result from the impacts fluid inertia on these two factors. We hypothesize that fluid inertia dramatically alters precipitation patterns by changing pore-scale 3D flow fields, which determine the mixing and surface accessibility of reactants.

To confirm this hypothesis, we conducted 3D reactive transport simulations to investigate the role of inertia and pore-scale flows on mineral precipitation. In the simulations of reactive transport, two reactants with a concentration of 1.5 mM are introduced from each inlet, as in the experiments. The flow field was obtained by solving the Navier–Stokes equations, while the concentration field was obtained by solving the advection–diffusion equation with a surface reaction term. We assume the surface reaction follows first-order kinetics from transition state theory (60), represented as $R = k(\Omega - 1)$, where k is the reaction rate constant and Ω is the saturation ratio defined as $\Omega = c_{\text{Ba}^{2+}} c_{\text{SO}_4^{2-}} / K_{sp}$, where K_{sp} refers to the solubility product of barite and equals to $1.1 \times 10^{-10} \text{ mol}^2 / \text{L}^2$ (61). The change in channel geometry caused by precipitation is not taken into account. Flow and concentration fields reached a steady state for all Re considered in this study, confirming that the flow regimes considered in this study belong to the steady inertial laminar regime (23, 30). We quantify two key parameters from the obtained concentration fields: the normalized saturation ratio at the top and bottom surfaces (Ω_{Surf}), and the projected, average saturation ratio of the bulk solution (Ω_{Soln}). To derive these parameters, we normalized both the surface saturation ratio and the projected, average saturation ratio against their respective maximum values observed at Re = 150. Homogeneous, solution reaction rates are proportional to bulk solution saturation ratios (Ω_{Soln}), while heterogeneous, surface reaction rates are dependent on surface saturation ratios (Ω_{Surf}). Hydrodynamics affects the surface reaction rate by influencing the surface saturation ratio. The Damköhler number ($Da = RSL/cu$, where S is the specific

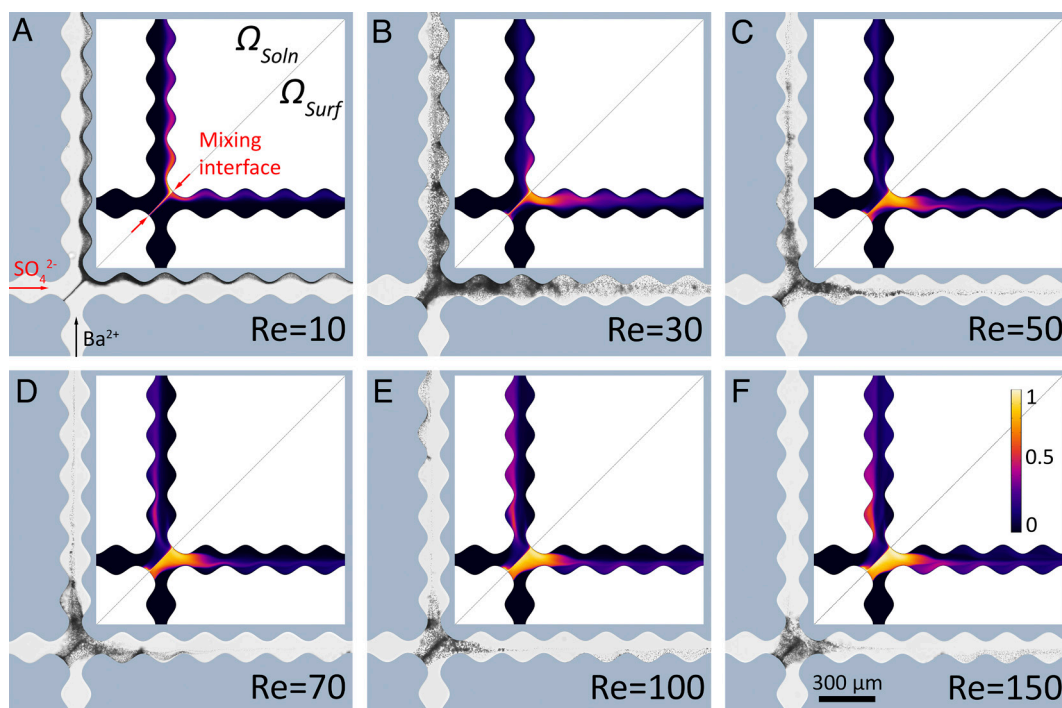


Fig. 1. The patterns of precipitation (shown in black), obtained from microfluidics experiments across varying Reynolds numbers (A–F). The barium chloride solution was introduced via the *Bottom* inlet while the sodium sulfate solution was injected from the *Left* inlet, maintaining a flow rate ratio of 1:1. Images were acquired at the end of the experiment with a total injection volume of 20 mL. The *Insets* provide the reactive transport simulation results, with the normalized projected bulk saturation ratio of the solution, Ω_{Soln} , shown in the *Upper Left*, and *Lower Right* shows the normalized saturation ratio at the surface, Ω_{Surf} . We normalized both the surface saturation ratio and the projected bulk saturation ratio with their respective maximum values at $Re = 150$. The colorbar in Fig. 1 F, *Inset* applies to both Ω_{Soln} and Ω_{Surf} . The red arrows indicate the mixing interface, aligned with the intersection's axis of symmetry. The precipitation patterns change dramatically with Re , accompanied by enhanced precipitation at intersections. The Ω_{Surf} maps match well with the precipitation patterns and are significantly different from the Ω_{Soln} map.

surface area of channel, L is the channel length, and c is the reactant concentration) is significantly smaller than 1 across all Re investigated. This indicates that the process is reaction-limited, with no significant change in reactant concentration occurring over the flow timescale. The normalized, spatial Ω_{Surf} and Ω_{Soln} are shown in the *Insets* of Fig. 1. The experimental precipitation patterns match very well with maps of Ω_{Surf} obtained from simulations for all Re values, as also evidenced by a quantitative comparison of the surface reaction area between experiments and simulations, as shown in *SI Appendix, Fig. S4*. In addition, maps of Ω_{Surf} differ from maps of Ω_{Soln} except for $Re = 10$, indicating inertia has distinct effects on surface reactions and solution reactions.

A recent study showed that inertial flows can induce reaction hot spots of homogenous, mixing-controlled reactions (31), which aligns well with the hot spots of Ω_{Soln} where enhanced saturation ratio values are found in the recirculating zones of the downstream channels (e.g., Fig. 1 F, *Inset*). However, the hot spots for surface reactions are mainly located at the intersection and show minimal overlap with Ω_{Soln} . The dramatic difference between Ω_{Surf} and Ω_{Soln} indicates that there may be strong 3D flows bringing solutes near the surface. Indeed, Dean flow and recirculation flow gradually emerge as Re increases (*SI Appendix, Fig. S5*). Dean flow refers to a secondary flow generated at intersections due to centrifugal force, manifested as two counterrotating vortices along the main flow direction (62), while the recirculation flow refers to the spiral and vortical motion of fluids within the flow separation zones. The simulation results suggest that the enhanced precipitation at channel intersections may be explained by the inherent 3D character of inertial flows.

3D Flow Topology Controls Precipitation Dynamics. As demonstrated in the previous section, inertial effects fundamentally alter precipitation patterns. To elucidate the specific mechanisms of how precipitation is controlled by fluid inertia, we plot the streamlines that travel close to the mixing interface, as shown in Fig. 2A. Specifically, the streamlines that pass through a plane that is parallel to and situated 1 μm away from the mixing interface (shown as gray panel) are visualized. These are reactive streamlines where the two solutions mix and result in precipitation. At $Re = 10$, the streamlines stay parallel to each other and are essentially two-dimensional (2D), with negligible z -directional flow. Thus, mixing, and subsequent precipitation, are focused along the mixing interface, resulting in the observed experimental precipitation pattern and elevated reaction rates at the mixing interface and downstream locations where reactive streamlines pass (refer to Fig. 1 A, *Inset* and Fig. 2A). As Re increases, the flow topology substantially changes, exhibiting pronounced 3D characteristics. At intermediate Re (e.g., $Re = 30$ & 50), the precipitation becomes much broader due to the emergence of secondary flow with strong z -directional flows that bring reactants to the channel surfaces, enhancing their surface accessibility. The reactive streamlines enter from the mid-depth of the channel (at $z = 0$), and they diverge toward the upper and lower surfaces along the mixing interface, evidenced by the change in streamline colors shown in Fig. 2A. Such diverging flow patterns are typified by Dean flow. At higher Re (e.g., $Re = 100$ & 150), strong 3D spiral and recirculating flows emerge near the inlet side of the mixing interface, and these 3D diverging flows further enhance mixing and reactant surface access, enhancing precipitation throughout the intersection. These effects magnify precipitation because flows slow down as they approach no-slip surfaces, further enhancing

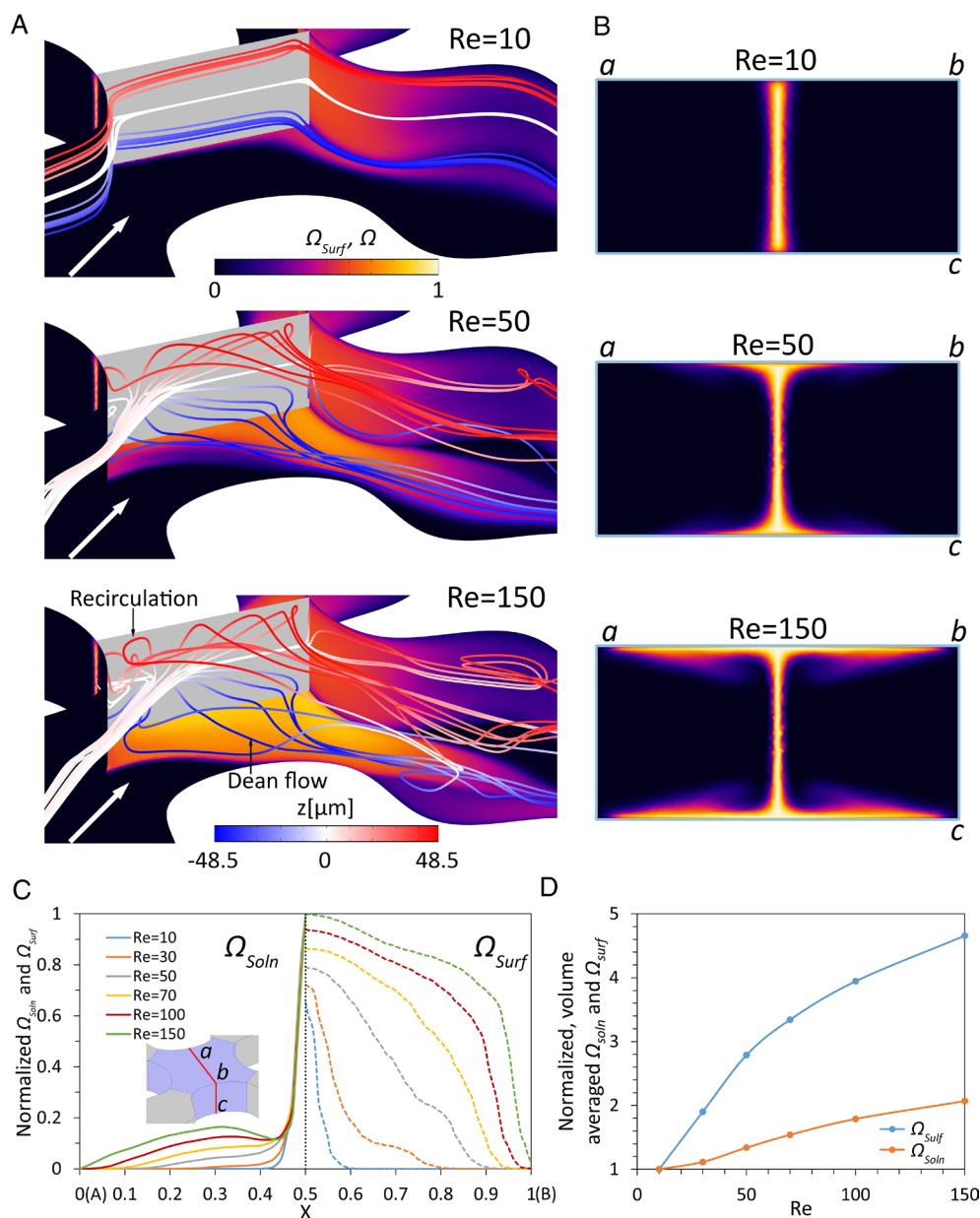


Fig. 2. 3D flow topology controls precipitation dynamics. (A) The 3D reactive streamlines passing along the mixing interface and Ω_{surf} near the intersection. The gray plane is the mixing interface. Only half of the intersection is shown here since the system is symmetric with respect to the mixing interface under even flow conditions. The white arrow shows the flow direction. At $Re = 10$, the flow is predominantly 2D. However, as the Re increases to 50, z -directional flows become evident. This 3D flow characteristic becomes even more pronounced at $Re = 150$. (B) The spatial distribution of Ω on the cut plane $a-b-c$. The reactants are localized at the mixing interface at $Re = 10$. As Re increases, the reactants begin to disperse away from the mixing interface, notably near the top and bottom surfaces. When Re is further increased to 150, the reactants spread across almost the entirety of the top and bottom surfaces. (C) The projected normalized saturation ratio Ω_{soln} ($X \leq 0.5$) and normalized surface saturation ratio Ω_{surf} ($X \geq 0.5$) along the cut line $a-b$. An increase in Re significantly increases Ω_{surf} along the cut line and only moderately affects Ω_{soln} profile. In the *Inset*, the intersection region and the cut plane are depicted in blue and red, respectively. (D) The normalized, volume-averaged Ω_{soln} and Ω_{surf} at the intersection as a function of Re . They are evaluated within the blue intersection region shown in Fig. 2 C, *Inset*. Inertia has a more pronounced effect on enhancing surface reactions than solution reactions at intersections.

potential nucleation and precipitation. While the reactive streamlines travel close to the upper and lower channel surfaces in the intersection, they eventually converge toward the mid-depth of the channel downstream of the intersection. The characteristics of reactive streamlines explain rapid, localized precipitation at the intersection. Strong mixing and enhanced surface accessibility lead to rapid precipitation at the intersection, but the Dean flow eventually brings reactants away from the surfaces (toward $z = 0$) and limits precipitation downstream. In addition to the changes in streamlines, the magnitude of vorticity and z -directional velocity increases significantly with increasing Re , accompanied by changes in the spatial patterns of vorticity and z -direction velocity maps.

This further demonstrates the emergence of 3D flows, which in turn lead to the change in precipitation patterns. Detailed information on vorticity and z -directional velocity can be found in *SI Appendix, Supporting Information Text* and *Figs. S6–S8*.

To further highlight the role of 3D flow topology on spatial reactant distribution, we present the spatial map of the saturation ratio, Ω , at a cross-section of the intersection (see plane $a-b-c$ in Fig. 2 C, *Inset*) in Fig. 2B. In the laminar flow regime ($Re = 10$), the reactants stay concentrated near the mixing interface, matching the streamline behavior. However, at an intermediate Re of 50, the 3D flows bring reactants toward the top and bottom surfaces. This trend becomes even more pronounced in the inertial

laminar flow regime at $Re = 150$, where the spatial distribution of reactants undergoes a significant transformation from the pattern observed at $Re = 10$. We observe elevated Ω values covering almost the entire top and bottom surfaces at $Re = 150$, a direct consequence of the inertia-induced 3D flow topology shown by the reactive streamlines. As Re increases, both reactant mixing and accessibility to surfaces at intersection zones increase, resulting in rapid and localized precipitation.

Quantitative insights into the distinct effects of inertia on solution and surface reactions are provided in Fig. 2C and D. Fig. 2C shows that the Ω_{Soln} is at maximum at the mixing interface ($x = 0.5$) and decays rapidly along the cutline $a-b$ for all Re values. Notably, an increase in Re does not substantially increase Ω_{Soln} in the intersection. In contrast, an increase in Re substantially increases Ω_{Surf} . This is due to the pronounced spatial heterogeneity in the reactant distribution at high Re , especially along the z -axis, with a high concentration of reactants primarily near the top and bottom surfaces (Fig. 2B). These observations demonstrate the distinct influence of inertia on solution versus surface reactions. We present the normalized, volume-averaged Ω_{Soln} and Ω_{Surf} at the intersection in Fig. 2D. These values are normalized with respect to the volume-averaged Ω_{Soln} and Ω_{Surf} at a Re value of 10. Additionally, the defined volume of the intersection, where these measurements were taken, is highlighted in blue in the *Inset* of Fig. 2C. As Re increases, an increasing trend is observed in both normalized, volume-averaged Ω_{Soln} and Ω_{Surf} . Yet, the rate of increase for the normalized, volume-averaged Ω_{Surf} is significantly larger than that of the normalized, volume-averaged Ω_{Soln} . This implies that inertia enhances surface reactions more than solution reactions at intersections, which is also evident from Fig. 1, *Insets*.

Inertia Effects Lead to Rapid Intersection Clogging. As fluid inertia greatly impacts precipitation, it will also have profound effects on the clogging of flow channels. To quantify the inertia effects on intersection clogging, we performed precipitation experiments over an extended period and estimated the time evolution of the channel hydraulic conductivity. Here, clogging refers to the progressive accumulation of precipitates that eventually lead to a significant reduction in channel hydraulic conductivity. The normalized hydraulic conductivity at time, t , is defined as $-K = \Delta p_0 / \Delta p_t$, where Δp_0 and Δp_t are the measured differential pressures of the channel at times 0 and t , respectively. The temporal evolution of $-K$ trends is dramatically different between the low and high Re , as shown in Fig. 3A. At $Re = 10$, conductivity decreased initially at a slow rate and then plateaued after 220 min of injection. The *Inset* experimental image shows the formation of a thin precipitate barrier at $Re = 10$, which blocked the mixing and led to sustained flow. Following the formation of a dense precipitation layer (at $t = 220$ min), a fluorescent tracer was introduced from one inlet to confirm the impermeability of the barrier and its function as a mixing barrier. As shown in *SI Appendix, Fig. S9*, no tracer was transported through the barrier, confirming its impermeability. In contrast, at $Re = 150$, $-K$ decreased much more rapidly to a value of 0.33 at $t = 73$ min, accompanied by significant clogging in the intersection and the channel throat. The experiment was stopped after 73 min due to the pressure reaching the upper limit of the pressure sensor. The significant clogging is evident in Fig. 3A, *Inset*, which shows that most of the intersection is filled with precipitates. These findings underscore that inertial 3D flows are a key determinant in intersection clogging: Lower Re conditions ($Re = 10$) lead to no clogging, whereas higher Re ($Re = 150$) facilitates rapid clogging.

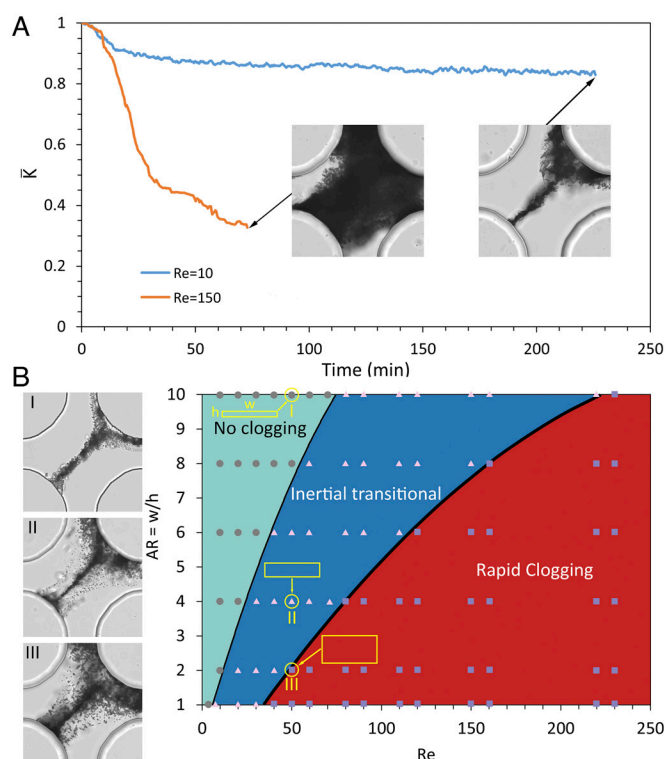


Fig. 3. Inertia effects lead to rapid intersection clogging. (A) The evolutions of normalized channel hydraulic conductivity with time for $Re = 10$ and 150 cases. The *Insets* show images of intersections at the end of the experiments, which is after injecting 38.9 mL for $Re = 10$ and 145 mL for $Re = 150$. A thin mixing barrier forms at $Re = 10$ leading to the precipitation shut-off, whereas the entire intersection is filled with precipitates at $Re = 150$. At $Re = 10$, these precipitates do not result in intersection clogging. However, at $Re = 150$, substantial clogging occurs. The precipitates are shown in black. (B) Experimental images show precipitation patterns at $Re = 50$ with AR values of 10, 4, and 2, respectively. The total injection volume for each experiment is 20 mL. The transition of clogging regimes from the no clogging (I) to inertial transitional (II) to clogging (III) regimes occurs as AR decreases. The phase diagram illustrates three distinct precipitation regimes as a function of Re and AR . Data points marked on the diagram are obtained from simulations. Yellow circles showcases that are shown in the *Left* experimental images.

Our findings demonstrate that inertia-induced 3D flows are the primary mechanism that determines the fate of intersection clogging. Previous studies have identified channel aspect ratio (AR), defined as the ratio of channel width to depth, as a crucial factor influencing the onset of 3D inertial flows, and should therefore impact clogging from mixing-induced precipitation (62). To further demonstrate the importance of inertia-induced 3D flow on mineral precipitation, we performed both microfluidic experiments and reactive transport simulations by varying both AR and Re . The three experimental images shown in Fig. 3B were performed at $Re = 50$ but with different AR s. The three images manifest markedly different spatial precipitation patterns with increasing AR corresponding to less clogging. As AR increases, the flow becomes more 2D and intersections are less likely to clog. This is further confirmed in simulation results plotted as a phase diagram of clogging regimes shown in Fig. 3B. Three regimes were observed, determined by the profile of spatial surface reaction rate along the pore throat that is downstream of the intersection (*SI Appendix, Fig. S10*). In the no-clogging regime (creeping flow regime), reactions are constrained to one side of the throat, and the extent of the reaction zone remains similar as Re increases. The rapid clogging regime is defined as the regime where the reaction rate has non-negligible values throughout the entire throat. This is a natural choice because the precipitation over the entire pore throat will lead to rapid clogging. The inertial

transitional regime is when the reaction zone is significantly larger than the no-clogging regime but does not cover the entire throat. It is noteworthy that the critical Re for the onset of rapid clogging increases with AR . This indicates that shallow channels with large AR are less prone to clogging than channels with AR closer to one – a finding that might seem counterintuitive at first. This is because the flow is more 2D when AR is large and a higher Re is required to generate inertial 3D flows in channels with a larger AR . In other words, at a given Re , the flow field becomes more 2D as the AR increases. When AR is one, the no-slip boundary conditions from all four sides interact, making 3D flows readily occur (62). As a result of this shift in flow characteristics, clogging regimes evolve from no clogging to inertial transitional, and eventually to rapid clogging as Re increases or AR approaches one. Interestingly, a previous study showed that channels with large AR values are more prone to clogging than those with small AR values during dissolution-precipitation processes of cement samples (63). This is likely attributed to the absence of fluid inertia, and the clogging behavior may become more complex at higher Re .

Flow Symmetry-Breaking Significantly Enhances Precipitation.

Natural and engineered fracture intersections are often characterized by uneven flow rates, irregular channel geometry, and multiple interconnected intersections that interact, departing from the relatively idealized symmetric conditions we have presented so far. Therefore, we performed further experiments and simulations to elucidate how symmetry-breaking affects the precipitation dynamics.

We first present the effects of symmetry-breaking on mixing-induced precipitation in sinusoidal channel intersections by increasing the

injection rate of sodium sulfate relative to the barium chloride flow rate (flow rate ratio of 1.2:1). Fig. 4A presents the precipitation patterns under this uneven flow condition at the end of each experiment. We explored inertial flow effects as before by changing the absolute injection rate between experiments while keeping the injection rate ratio at 1.2:1. Experimental videos that show the time evolution of precipitation patterns are provided in [Movies S3](#) and [S4](#). The simulated surface reaction rates are shown in [SI Appendix, Fig. S11](#), which closely captures the initial precipitation patterns. Interestingly, despite the higher injection rate of sodium sulfate solution, the precipitation occurs in both downstream channels. For uneven flow conditions, one might expect all the flow from the smaller injection rate side to enter one of the downstream channels, causing precipitation to only occur there. However, this is only the case for the early stage of small Re cases (e.g., $Re = 10$), and higher Re cases show simultaneous precipitation in both downstream channels. The precipitation patterns at the end of experiments for all Re values show significant precipitation in both downstream channels.

To provide a clearer picture of spatiotemporal precipitation dynamics in downstream channels, we quantify the changes in precipitation amount over time. We use grayscale values, corresponding to light blocked by precipitates, as a proxy for the total amount of precipitation. In Fig. 4B, we present the normalized grayscale intensity changes in downstream channels, $\Delta I / \Delta I_{\max}$, where ΔI is the total intensity variation as a function of injection volume, and $\Delta I_{\max}$ is the maximum intensity change after 3.27 mL of fluid injection in downstream channels. $\Delta I / \Delta I_{\max}$ should be proportional to the total amount of precipitation. The regions where the changes in normalized intensity are measured are shown in [SI Appendix, Fig. S12A](#). As shown in the first plot of Fig. 4B,

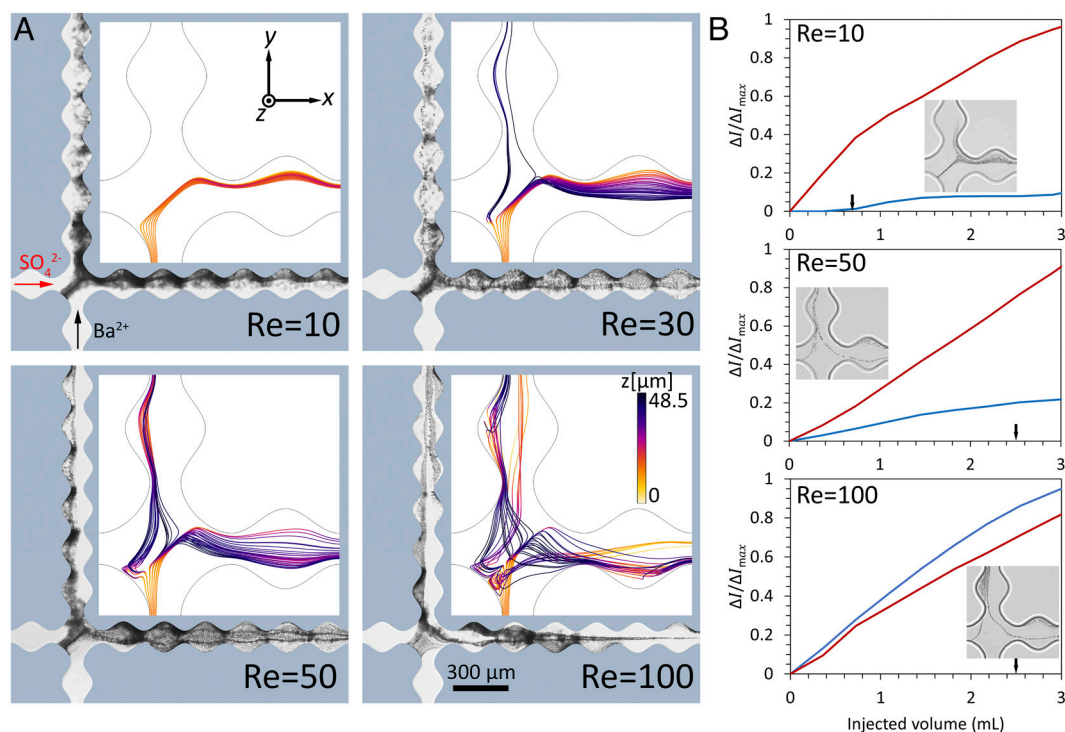


Fig. 4. Precipitation is significantly enhanced under uneven flow conditions. (A) Precipitation patterns at different Reynolds numbers under uneven flow conditions. Barium solution was injected from the bottom inlet and the sulfate solution was injected from the left inlet, with $Q_{\text{sulfate}}:Q_{\text{barium}} = 1.2:1$. Images were taken at the end of the experiment with a total injection volume of 20 mL. The *Inset* shows the streamlines enter both downstream channels at $Re \geq 30$, indicating the inertia-induced flow bifurcation. The color bar shows the z -location of the streamlines that are initiated from the *Bottom* inlet. Inertia-induced flow bifurcation significantly enhances precipitation in downstream channels along the y -direction. (B) The normalized grayscale intensity changes in the downstream channels as a function of injected volume. The red and blue lines are normalized precipitation amount along the x - and y -directional downstream channels, respectively. The black arrows indicate the time when the *Insets* were captured. Fluid inertia fundamentally changes spatiotemporal precipitation dynamics under uneven flow conditions. For $Re = 10$, no precipitation is observed in the downstream channels in the y -direction until the injected volume surpasses 0.72 mL. On the other hand, at $Re \geq 30$, precipitation occurs simultaneously in both downstream channels.

at $Re = 10$, the precipitation initially only occurs along the x-directional downstream channel, and no precipitation is observed in the y-directional downstream channel. As precipitation decreases the hydraulic conductivity of the x-directional channel, simultaneous precipitation occurs along both channels at around 0.72 mL of fluid injection. On the other hand, for $Re \geq 30$, the precipitation occurs simultaneously in both downstream channels despite the injection rate being larger from one side (see Fig. 4 B, *Insets*). Additionally, as Re increases, precipitation in the downstream channels shifts from one side of the channel to the opposite (e.g., compare precipitates along the y-directional downstream channel at $Re = 10$ and $Re = 50$ in Fig. 4A). The underlying mechanisms governing the observed precipitation patterns can be nicely explained again with the concept of reactive streamlines.

The reactive streamlines originating from the bottom inlet are shown in Fig. 4 A, *Insets*, and *SI Appendix, Fig. S12B* shows where the reactive streamlines are initiated. At $Re = 10$, the reactive streamlines all enter the x-directional downstream channel, but none enter the y-directional downstream channel. The streamlines exhibit minimal variation in their z-location, indicating predominantly 2D flow. However, for $Re \geq 30$, flow bifurcation is evident. Here, the streamlines from both upstream channels advectively enter both downstream channels (refer to Fig. 4 A, *Insets* and *SI Appendix, Fig. S13*), thereby enhancing simultaneous mixing-induced precipitation. Under even flow conditions, the streamlines are perfectly symmetric with respect to the mixing line, and streamlines cannot pass through the mixing interface, as shown in *SI Appendix, Fig. S5*. As a result, uneven flow conditions have much more widespread mixing and precipitation than even flow conditions. Reactive streamlines at higher Re cases exhibit pronounced changes in the z-direction, which indicates the strong 3D nature of the inertia-induced flow bifurcation. The darker, purple colors indicate locations near the top surface, and the bifurcated streamlines travel near top and bottom surfaces, increasing surface accessibility and thereby precipitation. According to these observations, inertial effects on precipitation enhancement are still valid under symmetry-breaking conditions and can have a more dramatic effect. Additionally, this implies that one can strategically engineer spatiotemporal precipitation patterns within channel systems by adjusting injection parameters.

Second, we sought to see how a rough channel geometry, which is common in natural and engineered systems, would influence the dynamics of inertial flows and mixing-induced precipitation via flow symmetry-breaking. This was accomplished through experiments in self-affine rough channel intersections ($H = 0.7$) with equal injection rates. Here, asymmetric flow fields result from rough channel walls. As shown in Fig. 5A, the primary observations are consistent in these rough channel intersections. A precipitate barrier is observed along the mixing interface at $Re = 10$, extending into downstream channels, a phenomenon also observed at sinusoidal channel intersections under both even and uneven flow conditions. As a result, precipitation shuts off and channels remain unclogged. In contrast, at $Re = 100$, precipitation occurs over the entire intersection. As a result of symmetry-breaking, inertia-induced flow bifurcation readily occurs, facilitating substantial increases in precipitation in both downstream channels. Furthermore, the rough surfaces of the channels substantially increase the Dean flow and flow recirculation (64). This significantly enhances fluid mixing, resulting in a widespread distribution of precipitation as well as local precipitation hot spots in downstream channels. Additional experiments were conducted to examine the effects of channel roughness on precipitation patterns. As illustrated in *SI Appendix, Fig. S14*, the primary precipitation patterns remained consistent despite changes in channel roughness,

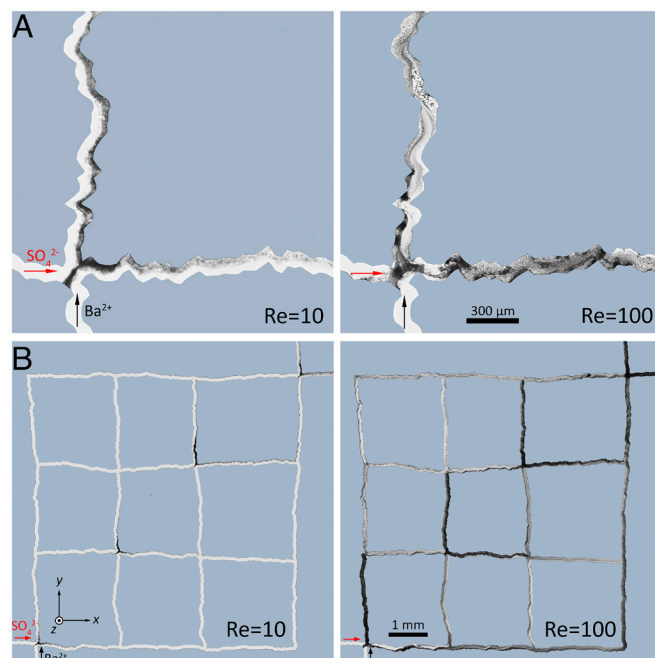


Fig. 5. Inertial effects on precipitation enhancement can be effectively translated from individual intersection scale to network scale. (A) Precipitation patterns at rough channel intersections for $Re = 10$ and 100. The total injection volume is 20 mL. A mixing barrier is formed by precipitates along the center of the intersection at $Re = 10$, whereas an enhanced precipitation at the intersection and downstream channels is observed at $Re = 100$. (B) Precipitation patterns at rough channel networks for $Re = 10$ and 100. The barium solution was injected from the bottom inlet and the sulfate solution was injected from the left inlet with a flow rate ratio of 1:1 for the rough intersection and network experiments. The total injection volume is 40 mL. At $Re = 10$, precipitation is predominantly localized at the intersections along the diagonal of the network. This localized pattern contrasts sharply with the scenario at $Re = 100$, where precipitation is extensively distributed across the entire network.

suggesting the generality of our findings. Thus, we can conclude that the precipitation pattern at intersections is primarily determined by fluid inertia and the nature of flow intersections (flow convergence followed by flow divergence) and not by the specifics of channel geometry.

Generalization to Rough Channel Networks. Finally, to confirm the effects of intersection-scale phenomena on network-scale precipitation, we conducted experiments in rough channel networks comprising 4×4 rough channels, as shown in Fig. 5B. We confirmed that the precipitation and clogging mechanisms observed at a single channel intersection are valid at the channel network level. Experiments in the flow networks show that mixing barriers form at low Re terminate precipitation without significant clogging, whereas fluid inertia at high Re dramatically enhances precipitation, leading to widespread precipitation and rapid clogging at the scale of channel networks. At $Re = 10$, precipitation occurs mostly at the mixing interfaces of the channel network at diagonal intersections. The formation of thin precipitate barriers at these intersections eventually terminates the reaction, without significant clogging. This is evidenced by the trend in normalized channel hydraulic conductivity shown in *SI Appendix, Fig. S15*. Initially, network-scale hydraulic conductivity decreased slightly and then soon plateaued to values above 0.8. In stark contrast, at $Re = 100$, inertial-induced flow bifurcation and recirculating flows significantly enhance fluid mixing and precipitation at the intersections, as well as throughout the entire network. Flow is uneven in nature in rough channel networks. This leads to rapid and severe clogging with normalized channel hydraulic conductivity decreased to 0.17 at $t = 16$ min.

Further decreases in channel hydraulic conductivity could not be captured because the pressure exceeded the maximum detection limit of the pressure sensor. [Movie S5](#) shows the full spatiotemporal dynamics of precipitation. More dynamic precipitation can be observed at $Re = 100$ compared to at $Re = 10$. At $Re = 100$, precipitation initially occurs primarily at the intersections and channels along the diagonal of the network. However, as the system becomes increasingly clogged, the pressure field evolves, leading to the activation of new flow paths. This can result in precipitation occurring in areas where it was not initially active.

Crystal morphology can play an important role in affecting precipitation dynamics. To further test the generality of our findings, we conducted additional experiments in rough channel networks using calcite as the target precipitate. Calcite is a common mineral observed in many processes including carbon mineralization. It is important to note that the morphology of calcite differs from that of barite. Barite and calcite belong to different crystal systems, with barite in the orthorhombic and calcite in the trigonal. In a similar manner to the barite precipitation experiments, we inject calcium chloride and sodium carbonate solutions with a concentration of 6 mM through two inlets into the rough channel networks. [SI Appendix, Fig. S16](#) and [Movie S6](#) show that the precipitation patterns and dynamics of calcite match those of barite, showing widely distributed, network-scale precipitation at $Re = 100$. This confirms that our findings are not limited to a specific crystal morphology.

Discussion

The study presented herein sheds light on the dynamics of fluid flow and its consequential impact on mineral precipitation, a phenomenon of profound significance in various subsurface and industrial processes such as carbon mineralization, oil recovery, geothermal energy production, and aquifer storage and recovery (19, 65–67). Despite burgeoning research underscoring the influence of fluid flow on mixing and chemical reactions (31, 68, 69), a mechanistic understanding of how fluid dynamics shape mixing-induced mineral precipitation and clogging has been limited. The inertial laminar regime, in particular, is increasingly recognized as an important flow regime in porous and fractured media (31, 64, 70). Mineral precipitation in an inertial laminar regime, however, has remained an open question.

To our knowledge, this investigation pioneers in revealing the effects of fluid inertia on mixing-induced mineral precipitation. At small Re ($Re < 10$), where inertia effects are negligible, precipitation manifests as a thin mixing barrier, consistent with previous pore-scale studies (42, 43, 71). This barrier curtails fluid mixing, thereby halting further precipitation without significant clogging. The lack of 3D flow at low Re is the main reason behind the termination of reactions and a marginal reduction in hydraulic conductivity. In contrast, at elevated Re values, the fluid inertia substantially alters the flow topology. The inertia-induced 3D Dean flow combined with recirculating flows enhances reactant surface access, thereby promoting precipitation at the intersections. Consequentially, precipitation occurs throughout the entire intersection at $Re > 50$, causing rapid clogging. The importance of 3D flows is further highlighted by showing the substantial influence of channel aspect ratio on clogging behavior. At a constant Re , an increase in AR suppresses the onset of 3D flows thus maintaining a non-clogged status. The findings are bolstered by extensive numerical simulations that establish a phase diagram of clogging regimes as a function of Re and aspect ratio. Inertia effects are shown to be more dramatic under symmetry-breaking conditions due to inertia-induced flow bifurcations that lead to

significant precipitation enhancement in the downstream channels. Finally, our observations at the single intersection scale effectively translate to a network scale, demonstrating consistent trends in precipitation behaviors across spatial scales.

Our findings suggest the potential to engineer spatiotemporal precipitation patterns and to predict clogging timescales using critical flow and channel geometry-related parameters. Depending on the application, one may be able to either minimize clogging by inducing 2D flows or maximize precipitation across fracture networks by promoting 3D flows. For subsurface applications necessitating prolonged fluid injection and opened flow paths, like enhanced geothermal systems, shale oil and gas recovery, and aquifer storage and recovery, operations should aim to avoid significant precipitation and clogging. Based on the results here, this could be accomplished by first preinjecting fluids at low flow rates to establish mixing barriers at flow intersections throughout a fracture network before ramping up flow rates to achieve target injection rates. Conversely, in processes where precipitation is beneficial, such as carbon mineralization, where mixing of CO_2 -charged water with divalent cation-rich formation water is known to contribute to the mineralization (72, 73), fluid injection conditions could be tailored to maximize precipitation across fracture networks, utilizing inertia effects. This study also has implications for dissolution and precipitation resulting from fluid–rock interactions, as fluid inertia will exert a strong impact on solute concentrations along fluid–rock interfaces, thereby controlling effective reaction rates. The effects of fluid inertia on coupled dissolution and precipitation processes will be an important area of future research. Last, this work presents important and generalizable insights into the influence of inertia versus the Péclet effect on precipitation. Without inertial effects, the increase in the Péclet number results in a sharpening of mixing interface and a thinning of the precipitation barrier. In stark contrast, fluid inertia significantly increases the precipitation area through the initiation of 3D flow topology.

In summary, this research underscores the important role of fluid inertia in dictating mixing-induced precipitation, with far-reaching implications for the manipulation of precipitation and hydraulic properties in flow network systems. As subsurface systems gain recognition as potential solutions to water, energy, and climate change challenges, understanding mineral precipitation becomes increasingly important. The contributions of this study are thus timely and have direct relevance to key subsurface applications.

Materials and Methods

Microfluidic Experiments. Soft photolithography is employed in the fabrication of microfluidic chips. Initially, a silicon wafer was spin-coated with SU-8 photoresist, creating a uniform layer with a specified depth. Subsequently, the silicon wafer was exposed to UV light and developed to yield a master mold for casting polydimethylsiloxane (PDMS) microstructures. Finally, the PDMS layers were bonded to glass slides by oxygen plasma treatment, creating microfluidic devices for conducting experiments (74). The dimensions of the orthogonally intersected sinusoidal channels are 1 cm x 1 cm in length and width, with three different depths of 20, 51, and 97 μm (Fig. 3B). The sinusoidal channel is periodic along the flow direction and is generated using the equation $y = \pm \frac{1}{2}[\Delta + (y_l - \Delta)\sin^2(\frac{\pi}{x_l}x)]$, where Δ is the pore throat size, y_l is the maximum size of channel cross-section, and x_l is the wavelength of the channel (75). Here, Δ , y_l , and x_l equal to 70, 180, and 240 μm , respectively. For visual reference, a schematic of the sinusoidal channel intersection is provided in [SI Appendix, Fig. S1](#). To extend the generality, we incorporated rough channel intersections and networks. Channel width (aperture) is constant at 100 μm , and channel depths are 83 and 95 μm , respectively. We generated the self-affine

surface with the SD of the height difference Δz between two points separated by a distance Δx (40 μm) given as $\sigma_{\Delta z} = \lambda^{-H} \sigma_{\Delta x}(\lambda \Delta x)$, where H is the Hurst exponent characterizing the surface roughness. To create the channels with a roughness representative of that found in nature and engineering systems, the value of H was set to 0.7 (76, 77). Experiments with $H = 0.6$ and 0.8 are also conducted and presented in [SI Appendix](#).

The concentration of barium chloride (c_{Ba}) and sodium sulfate solutions (c_{SO_4}) is 1.5 mM. We used high-purity water ($\geq 18\text{ M}\Omega$ resistivity) for preparing all solutions, and the chemicals were sourced from Fisher Scientific. The mix of the two solutions at the ratio of 1:1 gives the saturation index, $SI = \log \Omega$, where Ω is the saturation ratio, of 3.27, calculated using PHREEQC Interactive (78). The selection of this particular SI is driven by two key factors. First, it is representative of SI frequently encountered in both natural and engineering processes (79, 80). Second, the chosen SI leads to rapid precipitation, a characteristic crucial for conducting experiments within practical timeframes. Additionally, this rapid precipitation process is primarily governed by heterogeneous precipitation rather than homogeneous precipitation (48).

To ensure precise and controlled delivery of reactants, we utilized two pulsation-free syringe pumps (neMESYS 290N, Cetoni), which realized the coinjection of barium chloride and sodium sulfate solutions into the microfluidic chips at a consistent flow rate. An in-line pressure sensor was connected to the inlet tubing to measure pressure to determine the changes in fracture hydraulic conductivity (MPS3, Elveflow). We systematically varied the Re from 10 to 150 to investigate the inertia effects on the precipitation dynamic. We performed the experiments under both even (equal injection rate from both inlets) and uneven flow conditions to account for the complex flow conditions. To obtain detailed brightfield images that capture the precipitation process, we employed an inverted fluorescence microscope (Ti2-E, Nikon), equipped with CFI Plan UW 2 $\times$ and CFI Super Fluor 10 $\times$ objectives. For green fluorescence, fluorescence filters of peak excitation and emission of 470 and 525 nm, respectively were used. The microscope setup was complemented by a high-speed digital CMOS camera (ORCA-Flash4.0, Hamamatsu), which was set to a shutter speed of 30 ms. All the experiments were conducted at a room temperature of 22 $^{\circ}\text{C}$ and atmospheric pressure.

Reactive Transport Simulations. We performed steady-state 3D reactive transport simulations using COMSOL Multiphysics V5.5 to gain insights into the effects of inertia on flow, mixing, and reaction. A finite element method is used in COMSOL to solve the partial differential equations describing the reactive transport process. We consider an incompressible flow with a Newtonian fluid. The flow field was acquired by solving the continuity equation and Navier-Stokes equations through direct numerical simulation using the finite element method. The equations are,

$$\nabla \cdot \mathbf{u} = 0, \quad [1]$$

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$$\frac{\partial \mathbf{u}}{\partial t} + \nabla(\mathbf{u} \cdot \mathbf{u}) = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{g}, \quad [2]$$

where $\mathbf{u}$ is the fluid velocity, t is the time, ρ is the fluid density, p is the fluid pressure, ν is the kinematic viscosity of fluids, and $\mathbf{g}$ is the gravitational acceleration. The concentrations of barium and sulfate are assumed to be low enough such that the density and viscosity of the liquid are constant and equal to that of water. The temperature is also set to be constant and equal to 22 $^{\circ}\text{C}$. The solute concentration field was obtained by solving the advection-diffusion equation,

$$\frac{\partial c}{\partial t} = D \nabla^2 c - \nabla \cdot (c \mathbf{u}), \quad [3]$$

where c is the solute concentration. Because of the chosen experimental SI of 3.27, we consider that precipitation only occurs via heterogeneous nucleation on the solid surfaces of the channel. The boundary layer around the geometry surface was refined and contained six prism layers (81). The mesh sensitivity test confirmed that the mesh was fine enough for adequately simulating the surface reaction. The surface reaction rate is quantified by a first-order kinetic model (82),

$$D \nabla c = k(\Omega - 1), \quad [4]$$

where k was determined to be $10^{-7.62} \text{ mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$ from previously reported batch barite precipitation experiments (83), $\Omega = c_{Ba} c_{SO_4} / K_{sp}$ is the saturation ratio, and K_{sp} refers to the solubility product of barite.

The simulated domain has dimensions of $1,470 \times 1,470 \times 97 \mu\text{m}^3$ and a total mesh count of 5,016,650. Constant velocity boundary conditions were applied at the inlets, while constant pressure boundary conditions are imposed at the outlets, and the pressure is set to atmospheric. No-slip boundary conditions were used for the remaining channel walls. The geometry change due to precipitation was not considered in this study. Since the heterogeneous precipitation will be governed by reactant concentrations along the solid surfaces, what is important is the simulation accuracy of flow, solute transport, and mixing. The same setup was used and validated in our previous study (70), which shows that the simulation can accurately capture inertia-induced 3D flow and reactive transport.

Data, Materials, and Software Availability. Experimental data and numerical simulation setup have been deposited in Data Repository for U of M (DRUM) (84). All other data are included in the manuscript and/or [supporting information](#).

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